

04

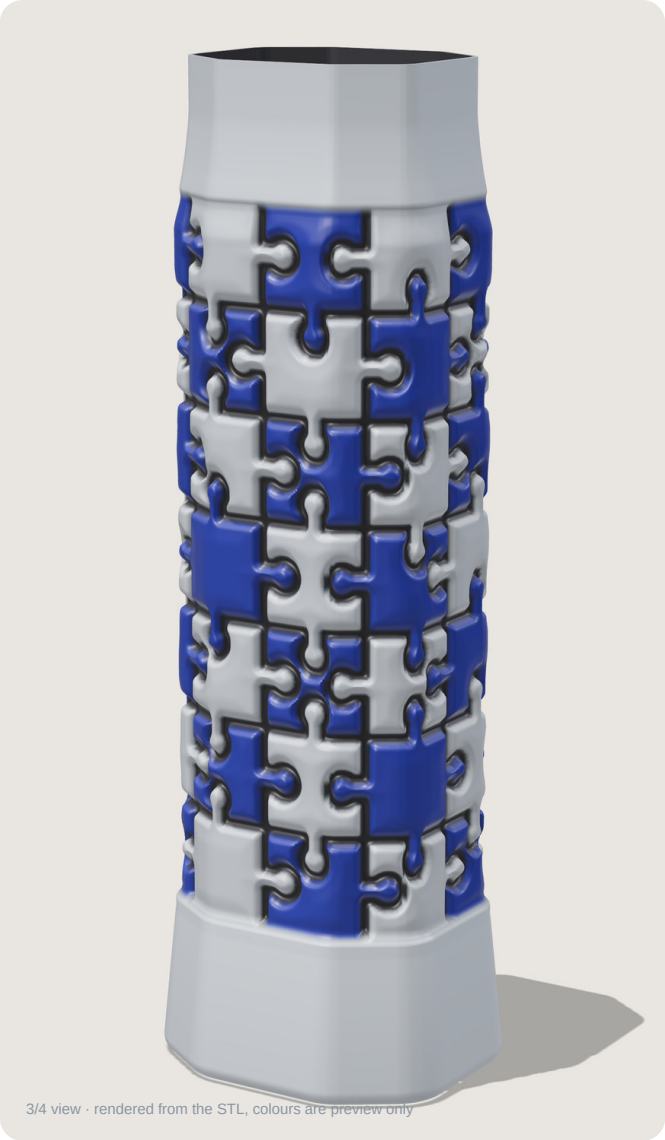
LOGIC-GRIP

Seamlessly interlocking ergonomic puzzle design.

48.0 g

PLA · 100 % infill · solid

✓ UNDER 50 G IN ALL 4 MATERIALS



AT A GLANCE

Overall height	132.82 mm
Max footprint (foot)	42.63 × 36.61 mm
Bore	32.06 × 26.04 mm, 127.8 deep
Textured band	z 21.5 – 116.0 mm (94.5 tall)
Outer size in band (max)	37.7 × 31.7 mm
Peak texture height	2.81 mm above bore
Base wall / min rim	1.2 mm / 0.8 mm
Volume / surface area	38.73 cm³ / 330 cm²

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³		48.0 g
PETG	1.27 g/cm³		49.2 g
TPU	1.21 g/cm³		46.9 g
ABS	1.04 g/cm³		40.3 g

50 g limit

COLOURWAY



3MF print palette

PRINT PROFILE

Infill / layer height	100 % / 0.20 mm
Layers	665
Orientation	upright, butt end down
Supports	none required
Filament, 1.75 mm	≈ 16.1 m (100 % flow)

DESIGN INTENT

Fifty-six jigsaw pieces tile the grip in alternating colours. Each piece is a shallow pillow separated by fine grooves and locked to its neighbours by mushroom knobs; knob directions are randomised so no two rows repeat.

Grooves are 0.84 mm wide, about two nozzle widths, and 1.6 mm deep.

VERIFIED AGAINST THE REFERENCE

Bore vs reference	identical
Outer envelope vs reference	+0.001 mm at most
Rim vs reference	+0.40 mm (deliberate, ≥ 0.8 mm wall)
Bodies / enclosed voids	1 / none
Material under 0.4 mm wide	0.000 % of volume
Longest bridge	none

KEY FEATURES

- 8 × 7 = 56 pieces in a checkerboard colourway
- Mushroom knobs: 2.9 mm neck, 4.2 mm head, 5.1 mm reach
- 0.84 mm grooves cut down to the base wall

FILES

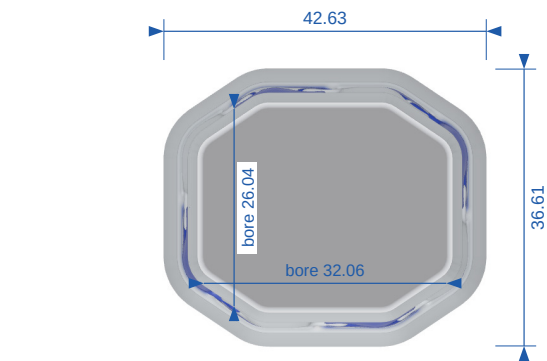
stl/04_logic_grip.stl	19.8 MB · 396,900 triangles
3mf/04_logic_grip.3mf	4.6 MB · colour

✓ FITMENT VERIFIED

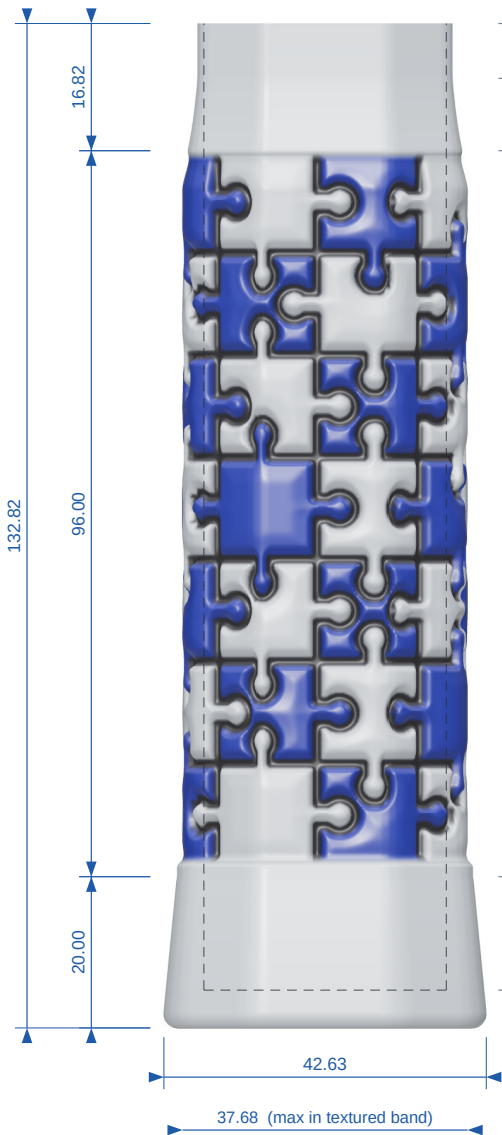
✓ SOLID · ONE CLOSED BODY

✓ PRINT-READY · 0.20 MM LAYERS

✓ 3MF VALIDATED



TOP VIEW (from +Z)

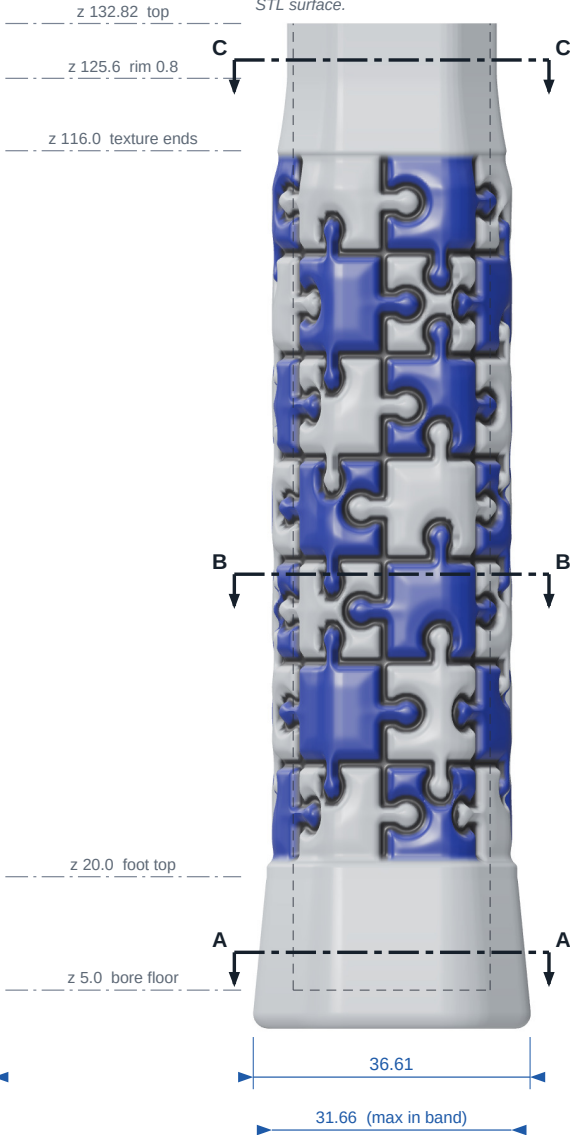


FRONT VIEW (from -Y)

KEY DIMENSIONS

Overall height	132.82
Overall width × depth (foot)	42.63 × 36.61
Bore width × depth	32.06 × 26.04
Bore depth (floor at z 5.0)	127.82
Outer W × D, textured band (max)	37.68 × 31.66
Foot top (z 20)	39.16 × 33.13
Butt face (z 0)	40.56 × 34.54
Peak texture height above bore	2.81
Base wall / minimum rim wall	1.20 / 0.80

All values in mm. Dashed lines: bore (hidden edges). Renders are the actual STL surface.



RIGHT VIEW (from +X)

A, B, C: see sheet 3

SCALE 1 : 1



PART

LOGIC-GRIP

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

- Views are true size: print this sheet at 100 %, do not fit to page.
- Dimensions are measured from the STL, not nominal.
- Bore, floor and outer envelope match the reference PickleballGrip_1.3mf.
- Rim is held at a minimum of 0.8 mm (reference tapers to 0.36 mm).

SCALE

1 : 1

MATERIAL

PLA / PETG / TPU

REVISION

2

UNITS

millimetres

INFILL

100 %

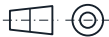
VOLUME

38.73 cm³

DATE

2026-09-30

PROJECTION



LAYER HEIGHT

0.20 mm

SURFACE AREA

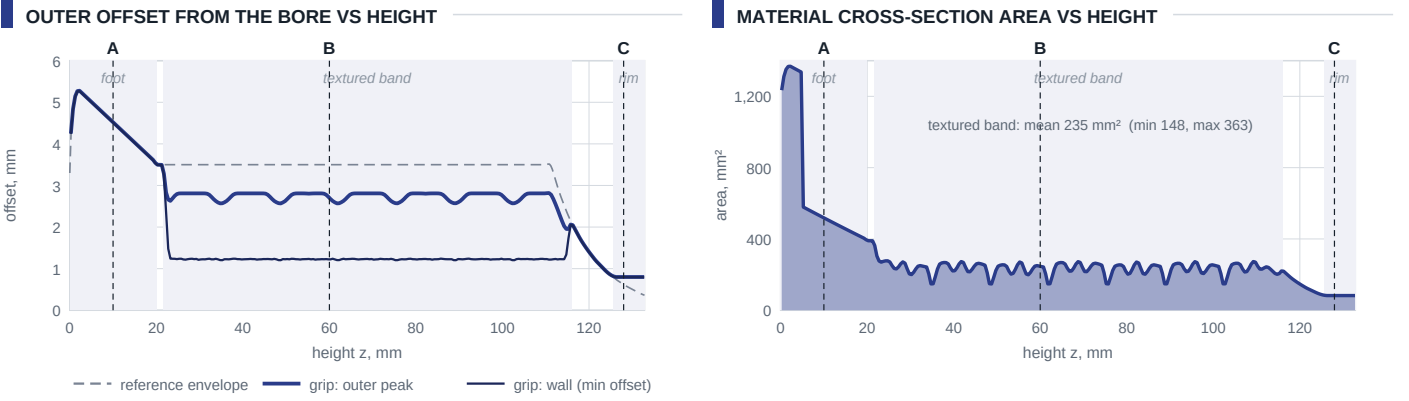
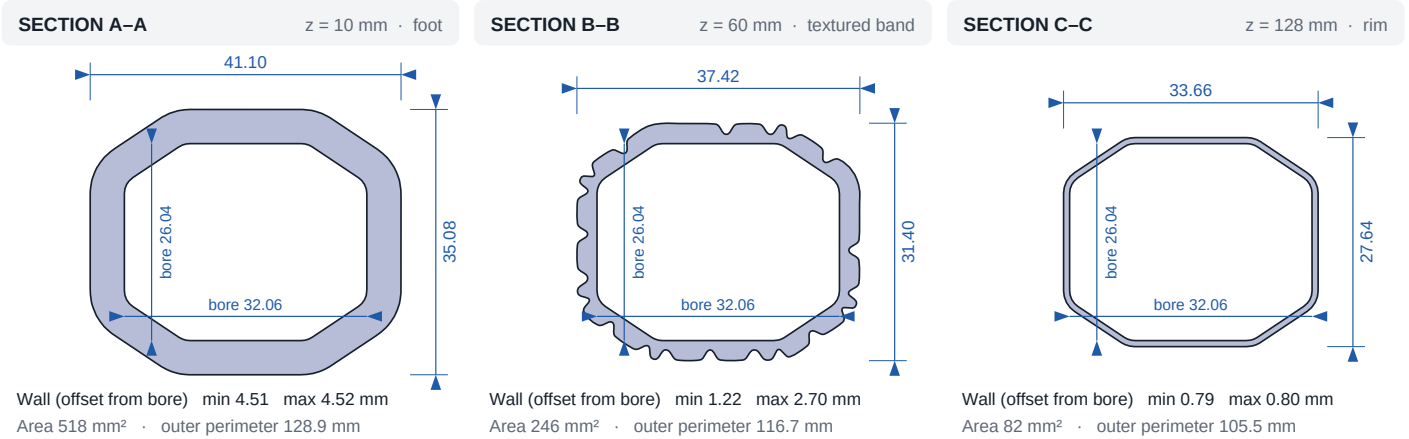
330 cm²

SHEET

2 of 4

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.



OVERALL & BORE

Overall length (Z)	132.82 mm
Overall width (X)	42.63 mm
Overall depth (Y)	36.61 mm
Bore width × depth	32.06 × 26.04 mm
Bore area / perimeter	751.0 mm ² / 100.5 mm
Bore depth (floor at z 5.0)	127.82 mm
Butt face (z 0)	40.56 × 34.54 mm
Outer W × D in band (max)	37.68 × 31.66 mm
Circumference in band	114.5 – 117.7 mm
Volume / surface area	38.73 cm ³ / 329.9 cm ²

ZONES BY HEIGHT

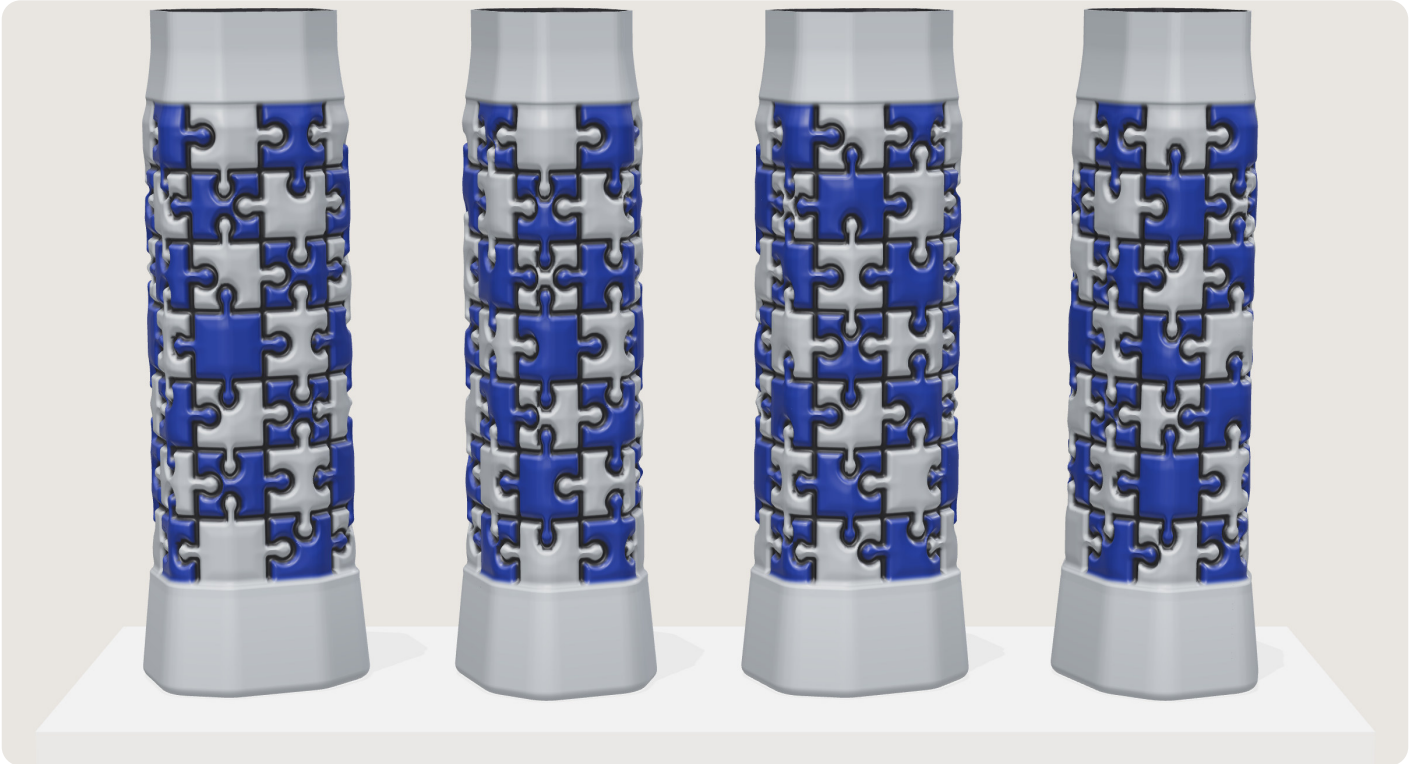
ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	5.0
Flared foot	0.0	20.0	20.0
Smooth collar	20.0	21.5	1.5
Textured band	21.5	116.0	94.5
Smooth neck (taper)	116.0	125.6	9.6
Rim (0.8 mm wall)	125.6	132.82	7.3
Base wall (design / measured min)			1.20 / 1.20 mm
Rim wall (design / measured min)			0.80 / 0.79 mm
Peak texture height above bore			2.81 mm

OUTER DIMENSIONS BY HEIGHT

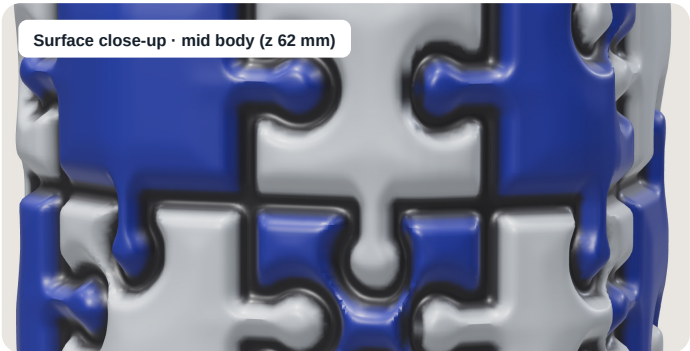
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM ²	WALL MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	958	5.00	5.01
10.0	41.10	35.08	128.9	518	4.51	4.52
20.0	39.11	33.09	122.6	393	3.52	3.52
30.0	37.57	31.60	115.1	202	1.22	2.81
45.0	37.59	31.65	117.1	254	1.23	2.81
60.0	37.42	31.40	116.7	247	1.22	2.70
75.0	37.23	31.19	114.9	179	1.21	2.58
90.0	37.24	31.27	115.3	208	1.23	2.61
105.0	37.56	31.50	117.2	262	1.23	2.75
116.0	36.17	30.15	113.4	220	2.04	2.05
125.0	33.77	27.75	105.8	88	0.85	0.85
132.0	33.66	27.64	105.5	82	0.79	0.80

Views & surface

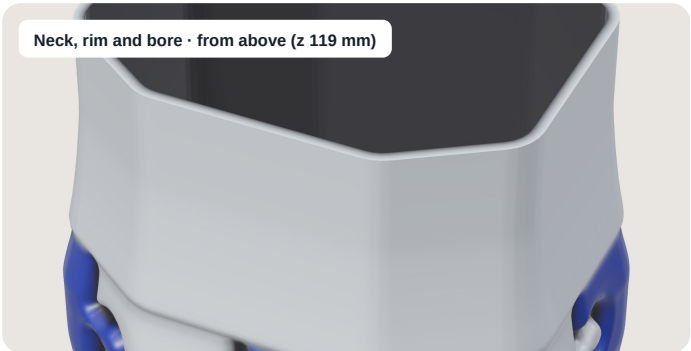
Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.



Four views, 90° apart · rendered from the STL, colours are preview only

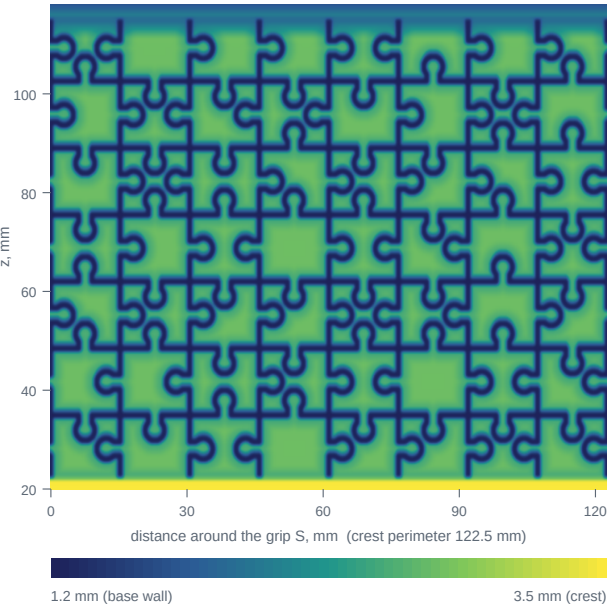


Surface close-up · mid body (z 62 mm)



Neck, rim and bore · from above (z 119 mm)

UNROLLED SURFACE · HEIGHT ABOVE BORE



DESIGN PARAMETERS

Pieces	8 × 7 = 56
Piece size	15.31 × 13.50 mm
Knob neck / head	2.9 mm / dia 4.2 mm
Knob reach	5.1 mm
Groove width	0.84 mm
Groove floor	1.20 mm (base wall)
Piece top height	up to 2.81 mm above bore

PRINT & VERIFICATION

Result: fitment / print-ready	PASS / PASS
Bodies · genus	1 · 0
Bore vs reference	identical
Envelope over reference	+0.001 mm
Material under 0.4 mm wide	0.000 %
Longest bridge	none